

University of Essex

MSc Artificial Intelligence

Module: KRAR_PCOM7E October 2025

Unit 1: Collaborative Discussion – Knowledge Representation Systems

Initial Discussion

The discussion at hand revolves around two questions, each addressed in turn:

1. Is Knowledge Representation (“KR”) a new development tied solely to its applications in computing?
2. Is KR useful without reasoning?

For the first question, my view is that KR as a term used in the computing context is necessarily or tautologically a recently-coined term on its face, but the aims of KR underpin and are shared with much of formal and symbolic logic as a philosophical field (Partarakis and Zabulis, 2024). In that sense, KR is part of a long history of inquiry into the subject (Partarakis and Zabulis, 2024; Poirier, 2024).

For the second question, my view is that KR likely does have some amount of inherent value when separated from any form of a reasoning component, but that value is limited. For example, a dictionary, or a collection of symbolic representations, of a particular subject matter area is useful in the sense of identifying those objects or persons or concepts that are important enough to be identified at all.

Without putting such a KR collection in context of a system of logical reasoning, however, the question of its utility cannot be answered (Bagal et al, 2024; Cambria, 2024). Knowing that symbols exist to identify one brand of car as different from another brand takes one as far as creating the awareness that different car brands exist, or are asserted to exist, but that awareness by itself does not provide any tools for evaluating the importance of that information. For example, some form of systematic reasoning is still needed to distinguish between brands that one would want to purchase, versus other brands that one would avoid purchasing.

In that sense, a KR corpus and some form of a logical reasoning system are inseparable, if they are to be put to practical use in real world scenarios (Bagal et al, 2024; Cambria, 2024).

References

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Peer Reponse No. 1

While your summary of the value of knowledge representation (KR) being connected to reasoning, I do think it is worth noting that there are many examples of KR used as a way to transfer knowledge to an LLM platform without the use of reasoning for functionality.

As an example, data frames in Python are a specific method of organizing and retrieving knowledge on a given topic, with the flexibility of defining what data is held and how it is labeled (De Ridder, 2025). They are not, though, designed for use in extrapolation for deductive reasoning. While there are multiple ways to use a data frame in conjunction with Python commands to manipulate contained data (Stojiljković, 2023), their primary purpose is to support large data sets while avoiding memory issues (Parulpandey, 2021).

That said, there has been research done to improve the use of data frames by introducing reasoning via prompt-based instructions. Gao, Chen and Sun (2025) suggested a “Learn then Retrieve” structure, called LRTab, to introduce reasoning capabilities based on tabular data. While this would greatly increase functionality based on this method of KR, it does highlight that this type of KR structure is not inherently built to leverage reasoning.

References

De Ridder, A. (2025) 'SmythOS - How intelligent agents use knowledge representation for Decision-Making,' SmythOS, 2 June.
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Gao, C., Chen, J. and Sun, J. (2025) Utilizing training data to improve LLM reasoning for tabular understanding. <https://arxiv.org/html/2508.18676v1>.

Parulpandey (2021) Speed up your Data munging with Python's Datatable.
<https://www.kaggle.com/code/parulpandey/speed-up-your-data-munging-with-python-s-datatable>.

Stojiljković, M. (2023) The pandas DataFrame: Make Working With Data Delightful.
<https://realpython.com/pandas-dataframe/>.

Peer Response No. 2

Your essay presents a well-argued, conceptually rich, and academically coherent discussion that effectively addresses both aspects of the prompt. One of its major strengths lies in your clear differentiation between Knowledge Representation (KR) as a recent computational development and as a concept deeply rooted in historical and philosophical traditions. By referencing contemporary research (Partarakis and Zabulis, 2024; Poirier, 2024), you successfully demonstrate that KR builds upon centuries of inquiry into logic, symbolism, and the formalisation of human understanding (GeeksforGeeks, 2024). This historical framing adds significant intellectual depth and demonstrates a critical awareness of KR's evolution beyond its technological applications (tutorialspoint, 2015).

Equally commendable is your balanced treatment of the relationship between KR and reasoning. You argue convincingly that while KR has standalone descriptive value, its true effectiveness lies in its integration with reasoning mechanisms that transform static representations into actionable knowledge (Bagal et al., 2024; Cambria, 2024). Your use of an accessible analogy involving car brands is particularly effective, as it grounds an abstract concept in a relatable context.

To further enhance the analysis, consider expanding on how contemporary Artificial Intelligence (AI) models—such as knowledge graphs or hybrid neuro-symbolic systems—operationalise this integration (Nawaz, Anees-ur-Rahaman and Saeed, 2025). Overall, this is a thoughtful, well-supported, and critically informed essay that demonstrates both depth of understanding and analytical maturity.

References

Bagal, V.C., Rane, A. L., Bhattacharya, D., Banubakode, A. and Mahalle, V. S. (2024). Knowledge Representation in Artificial Intelligence - A Practical Approach. Benthamdirect.com, [online] pp. 210–222. Available at: <https://www.benthamdirect.com/content/books/9789815179606.chapter-11>.

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tutorialspoint (2015). AI Knowledge Representation. [online] Tutorialspoint.com. Available at: https://www.tutorialspoint.com/artificial_intelligence/ai_knowledge_representation.htm.

Discussion Summary

My initial post made two assertions:

First, that knowledge representation ("KR"), taken at the highest level, is not a new concept or avenue of inquiry, but rather part of a long history of inquiry into the subject across many fields of research (Partarakis and Zabulis, 2024; Poirier, 2024).

Second, I asserted that without aligning any given KR system with a corresponding system of logical reasoning for analyzing and manipulating those same symbolic representations to reach actionable conclusions, such a KR system would have limited utility for real world applications (Bagal et al, 2024; Cambria, 2024). I made the comparison of a KR system that identified different car brands; a system of reasoning is needed to allow the knowledge that a set of car brands exists to be put to practical use in a real world scenario of purchasing a car.

Peer responses noted that:

First, a system of symbolic representations can be put to use to as a knowledge transfer mechanism independent of any logical reasoning system used to extrapolate conclusions from it; for example, by acting as a mechanism for transferring pre-categorized data sets between application platforms (Goa, Chen and Sun, 2025).

Second, contemporary artificial intelligence models in fact make use both of KR systems by themselves, as well as hybridized systems that combine symbolic representations with embedded reasoning systems (Nawaz, Anees-ur-Rahaman and Saeed, 2025).

References

Bagal, V.C., Rane, A. L., Bhattacharya, D., Banubakode, A. and Mahalle, V. S. (2024). Knowledge Representation in Artificial Intelligence - A Practical Approach. *Benthamdirect.com*, [online] pp. 210–222. Available at: <https://www.benthamdirect.com/content/books/9789815179606.chapter-11>.

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